

Physics of Semiconductor Devices

ECE210: PSD



INDRAPRASTHA INSTITUTE *of*
INFORMATION TECHNOLOGY
DELHI

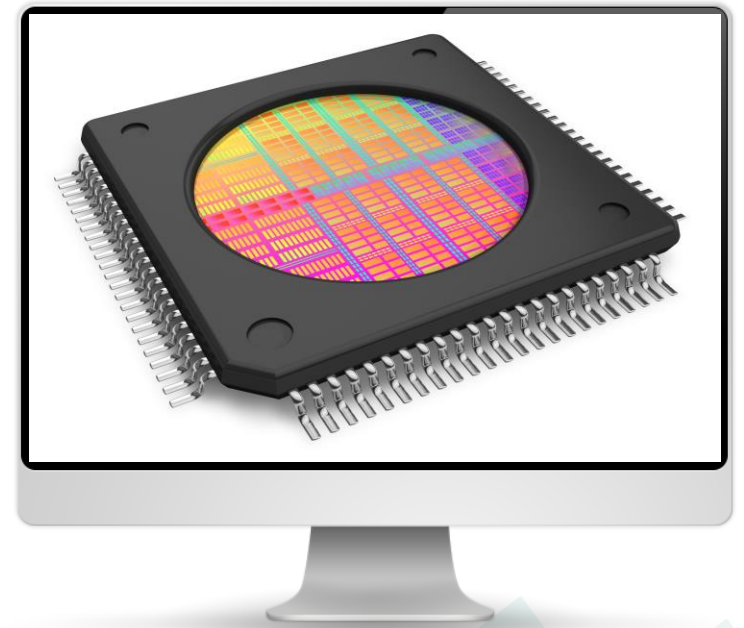
Dr. Ram Krishna Ghosh

week 01



**Semiconductor physics
& devices**

**Foundation of modern
electronics**



Mission and Vision of the Institute

The mission of IIT-Delhi is to be a global centre of excellence in Information Technology education, training and research. Its twin aims are:

- To carry out advanced research and development in information and software technologies, and in leveraging IT in specific domain areas.
- To train and educate, at both undergraduate and postgraduate levels, engineers of outstanding ability who can become innovators and new product creators.

The vision of the Institute is to be a world-class research-led institution of higher education in IT and allied areas which

- Is globally respected for research and education
- Offers thriving UG and PG programs
- Is socially relevant, industry facing, and globally connected

Mission and Vision of the ECE department

The **ECE Department's mission** is to carry out advanced research and development in various areas of Electronics & Communication Engineering with different application domains, and to train and educate, at both undergraduate and postgraduate levels, engineers of outstanding ability who can become innovators and new product creators.

The **vision of the department** is to be a world class department which:

- Is globally respected for its research capability, with some research groups being considered as among the leaders globally and within the country.
- Has thriving UG and PG programs.
- Is socially relevant, globally linked and industry facing.

About the Course



PSD: Objective



- To develop the fundamental concepts of semiconductor device physics, which are necessary to understand electronic devices.
- It will introduce students to the materials, transport mechanisms, and basic insight into various electronic and optoelectronic devices.
- To prepare for more advanced courses in semiconductor technology (e.g. analog, digital, mixed signal IC design, IC fabrication, Advanced SSD, Nanoelectronics, QMD, Memory Design, etc.).



PSD: Expected outcome



- Students will be able to explain the basics of semiconductor physics.
- Students will be able to describe the transport mechanism and current-voltage characteristics of pn-junction diode and Schottky diode.
- The students will be able to describe optoelectronic devices like solar cells, LEDs and their theoretical frameworks.
- Students will have a concept of classical MOSFETs and their characteristics.



PSD: Course plan



week 1, 2	Introduction – Basic quantum mechanics – Crystal Structure Importance of semiconductor devices and their applications, basics of semiconductor physics, Schrodinger's equation, crystal structure, Bloch's theorem, Free electron model. energy band formation, Energy dispersion relation, electron-hole pair generation, classification of semiconductors.
week 3, 4, 5	Band theory of solids – Semiconductor physics – Carrier transport Effective mass, Density of States, Intrinsic carrier density, Fermi level, doping and extrinsic semiconductors, charge neutrality relationship. Carrier transport: Diffusion current, drift current, Einstein relation, carrier scattering, mobility, , velocity saturation, generation and recombination of carriers.
week 6, 7, 8	Contacts & junctions – Semiconductor devices Types of junctions (ohmic, near ohmic, rectifying), carrier injection, pn junction, band bending & energy band diagram, pn junction electrostatics under equilibrium, fixed and free charge, depletion approximation, electric field at the junction, built-in potential, quasi-Fermi levels. pn junction under applied bias, band-bending under applied bias, Zener and avalanche breakdown.

PSD: Course plan



week 9	Physics of junction diodes Minority carrier injection and spatial variation in depletion and quasi-neutral regions, depletion capacitance in pn junction, ideal diode equation. Diode non-idealities, , Schottky diode, modifications of barrier potential in Schottky diode, Fermi level pinning.
week 10	Compound semiconductors – Photovoltaics Compound semiconductors, optical absorption, Solar cells - principle, J-V characteristics of solar cell, Shockley-Queisser limit, different types of solar cells
week 11	Light emitting devices Heterostructures, LEDs and their luminescent efficiency, double heterostructure and population inversion, diode lasers, L-I characteristics and threshold current.
week 12, 13	Foundation of MOS capacitor, MOSFET Working principle, I-V characteristics, pinch-off condition

PSD: Any Pre-requisite (Desirable)?



- A little bit of Solid State Physics,
 - Basic Electronics,
 - Electrostatics
- &
- Basics of Quantum Physics



PSD: Schedule, Office Hours, TAs



- ❖ Lecture: C02 (Old Academic Building, Ground Floor)
 - Monday and Thursday: 09:30 am – 11.00 am
- ❖ Instructor's Office Hours: B601 (R&D Building)
 - Tuesday and Thursday: 4:30 pm - 5:15 pm

TAs

Rouf Rahman Sheikh (Email id: roufs@iiitd.ac.in)

- Office : B616, Quantum Electronics Lab, R&D
- Office Hours : will be announced very soon

**Sai Spandana Thallam (Email id:
spandana22173@iiitd.ac.in)**

Join Google Classroom:

➤ Code : **eidlggr**

PSD: Reference Books



Text-book for this course:

- D. A. Neaman, *"Semiconductor Physics and Devices"*
- Ben G. Streetman and Sanjay Banerjee, *"Solid State Electronic Devices"*

Other reference books:

- C. L. Tang, *Fundamentals of Quantum Mechanics: For Solid State Electronics and Optics*
- S. M. Sze and K. K. Ng, *Physics of Semiconductor Devices*
- Robert F. Pierret, *Advanced Semiconductor Fundamentals (Modular Series on Solid State Devices, V. 6)*
- Jeffery L. Gray, *The Physics of the Solar Cell (Chapter 3)*
- E. Fred Schubert, *Light-Emitting Diodes*

PSD: Evaluation Criteria



Type of Evaluation	% Contribution in Grade
Assignment	20
Quiz	20
Mid-sem	30
End-sem	30



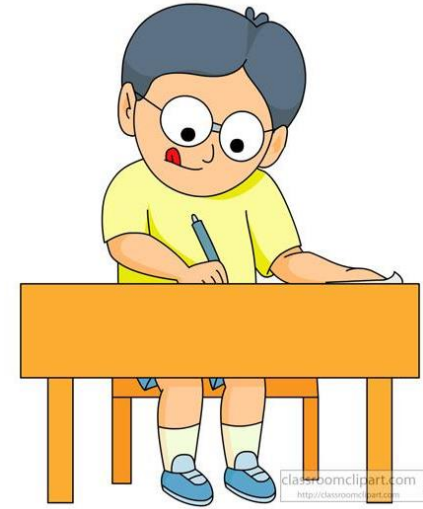
- Grading will be **relative**
- Less than 30% in absolute term: **FAIL grade**
- Quiz, mid & end sem will be closed book exam.
- **V. V. IMP: All exam questions will be based on class lectures, class participations.**

PSD: Assignments



2 Assignments to be done individually

- Final score will be scaled to 20%
- Answer should be hand-written on A4 size paper and submitted in Google classroom (no printout submission allowed).



	Assignment 1 (Tentative Dates)	Assignment 2 (Tentative Dates)
Assignment Upload	5th Feb	8th April
Deadline	15th Feb	18th April

Be punctual in online submission

- Each day of delay will attract 20 % penalty and after 3 days delay no more submission
- No excuse of internet down or only a few seconds of delay (Try to upload well in advance)

PSD: Quizzes



Dates for quiz will not be announced [[Surprise quizzes!](#)]

- 4 or 6 Quizzes of Equal Weightage [2 or 3 before mid sem & 2 or 3 after mid sem]
- **No retakes under ANY circumstance**
- Final score will be scaled to 20%
- Bring your own CALCULATOR
- Closed book exams, [if required, I will share One A4 sheet](#) in quiz, mid and endsem with complex/lengthy formulae which is difficult to memorize.
- No sharing of notes/calculator will be allowed

PSD: Attendance Policy



Student with less than 30% attendance [less than 8 classes] will be awarded FAIL grade *at the discretion of the instructor* [even when the total marks is above the PASS grade]

ATTENTION: DO NOT Register for the course if you do not plan to attend the classes regularly (for ANY reason!)

SSD: Policies



- Be on time (no late entry will be allow)
- Bring your own calculator
- Keep mobile phone, laptop in silence mode and inside your bag (should be switched off)
- Do not [contain a discussion among your peers](#),
- let the rest of the class know what are your concerns (may be some other students have similar concerns too!)
- Do not move around in classroom during lecture
- Always fill the seats from the front rows
- IIIT Delhi [plagiarism policy will be strictly followed](#) in this class



IIITD plagiarism policy



	Misconduct/use of unfair means in assignments/projects	Penalty
1.	First misconduct/ use of unfair means during the entire stay in IIIT-Delhi	Zero in the assignment (awarded by faculty) + one letter grade less in the course.
2.	Second misconduct/ use of unfair means during entire stay in IIIT-Delhi	Student is assigned an F grade in the course.
3.	Third or further misconduct/ use of unfair means during the entire stay in IIIT-Delhi	Student is assigned an F grade in the course and the case is reported to DAC, who may suspend the student for 1 semester to a year.

<https://www.iiitd.ac.in/sites/default/files/docs/education/2017/Plagiarism%20Policy%20-%20Updated.pdf>

IIITD plagiarism policy



	Misconduct/use of unfair means in quiz/midsem/endsem	Penalty
1.	First misconduct/ use of unfair means during the entire stay in IIIT-Delhi	Student is assigned an F grade in the course.
2.	Second misconduct/ use of unfair means during the entire stay in IIIT-Delhi	Student is assigned an F grade in the course and student may be suspended from the program for 1-2 semesters by DAC.
3.	Third misconduct/ use of unfair means during the entire stay in IIIT-Delhi	Student is assigned an F grade in the course and student's program may be terminated by DAC.

<https://www.iiitd.ac.in/sites/default/files/docs/education/2017/Plagiarism%20Policy%20-%20Updated.pdf>

SSD: Questions/Doubts



1. Class: Many may have the same questions
2. Just After Class
3. Office Hours
4. During revision
5. No last minute query for assignment submission
 - Clear all your doubts at least 2 days before submission

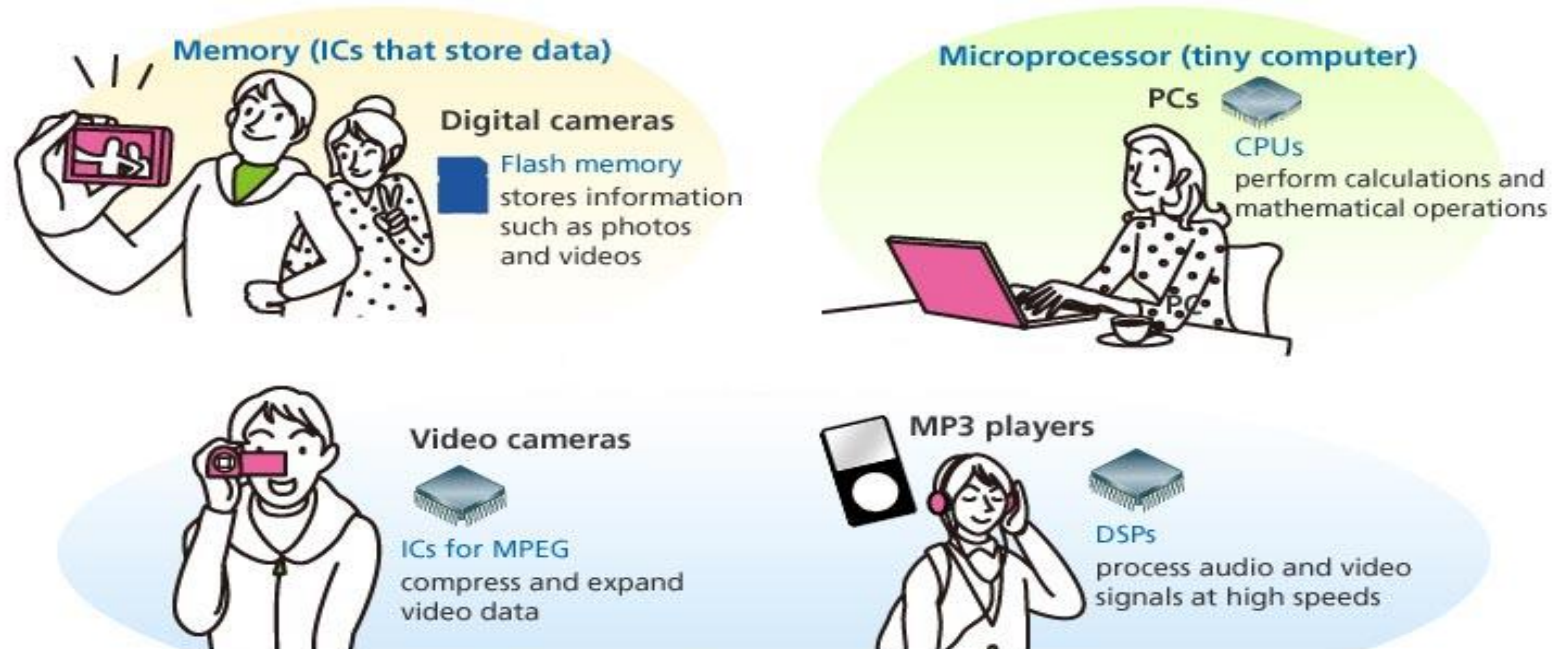


Expectation

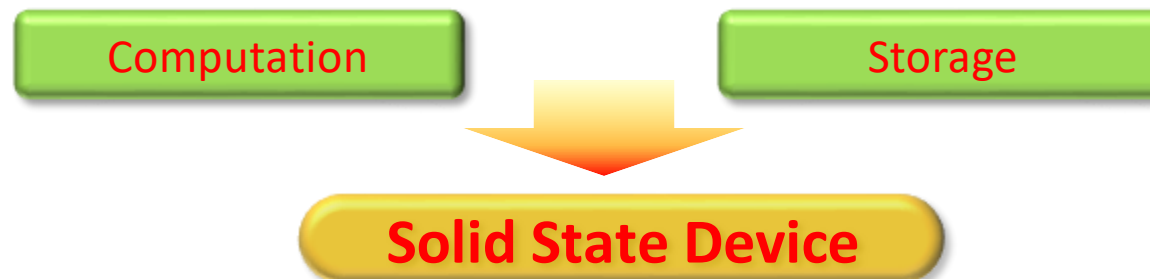
- Revise last day lecture before coming
- Queries will be cleared at the start of every lecture
- Ask questions and be active participants
- Give feedback



Electronic devices and applications



any application on an electronics gadget consists of two functions



stores and manipulates the computational state variable among two or more allowed state

What is solid-state electronics?



- Electronic action of devices occurred in a solid-state (as distinct from vacuum tubes)
- Practically, it means semiconductor-based electronics
- Took off very fast after the invention of semiconductor-based transistor

Invention of the point-contact transistor

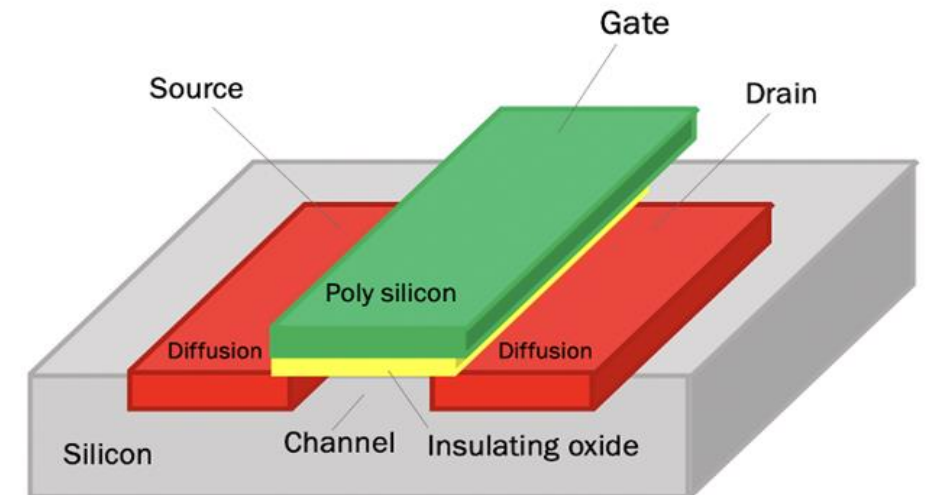


Walter Brattain (right), John Bardeen (left), and William Shockley (center), Bell Labs, 1947



Nobel Prize in Physics 1956

Modern MOS Transistor



Semiconductor technology hierarchy

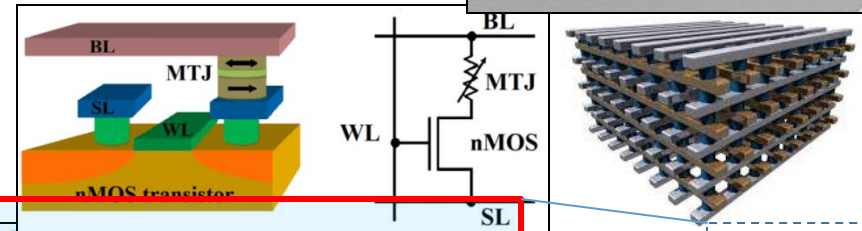


Any digital gadget needs the evolution of all these directions for future electronics

Application

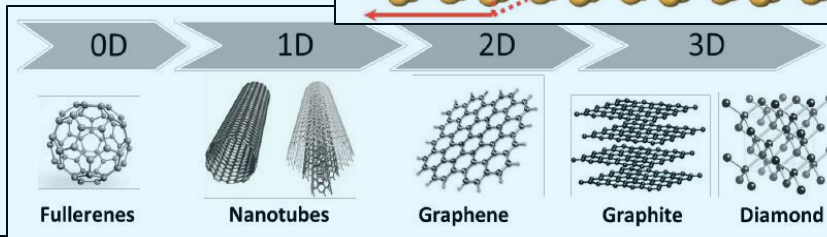


Circuit, architecture & integration

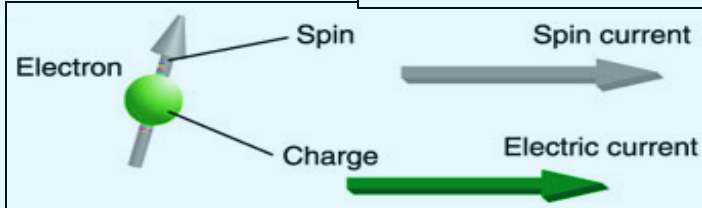


device

material



physics



devices

ECE210

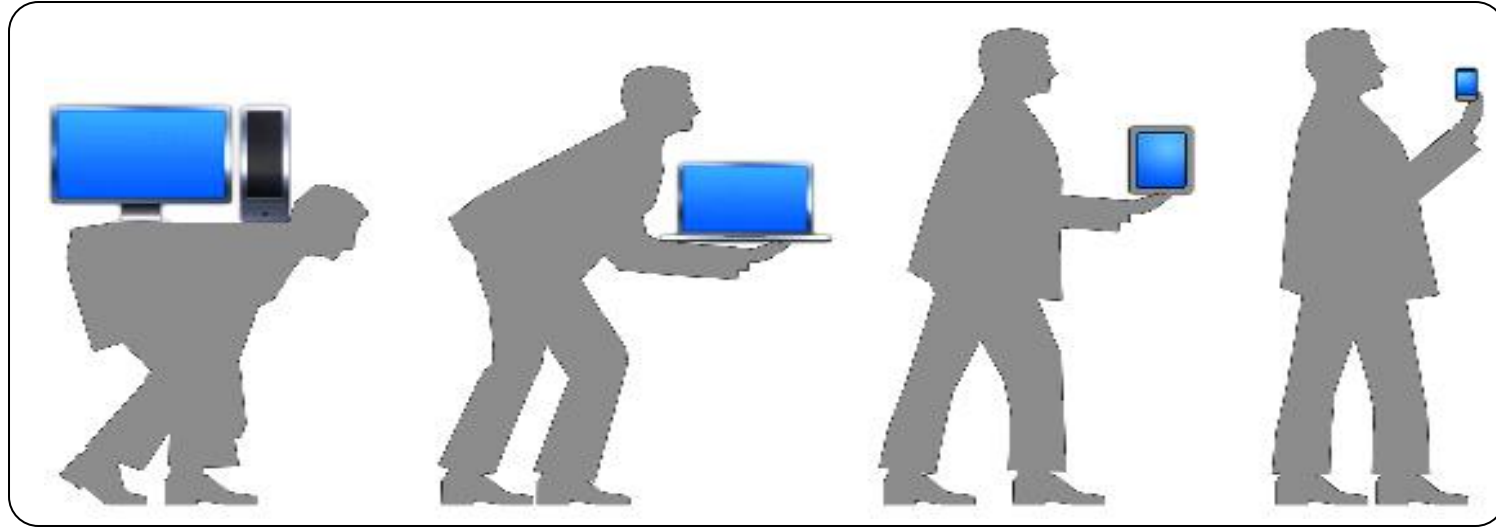
physics

materials

Miniaturization of electronic Devices



Why do we need to go to the "smaller" scale ??



Why should you care?

Incredibly exciting interdisciplinary area with a vast economic and business impact, and much need for physical understanding.

Scaling of devices: Moore's law



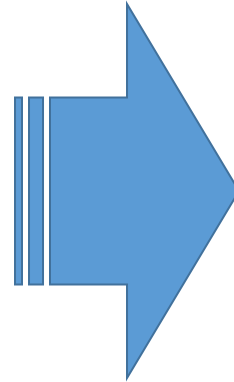
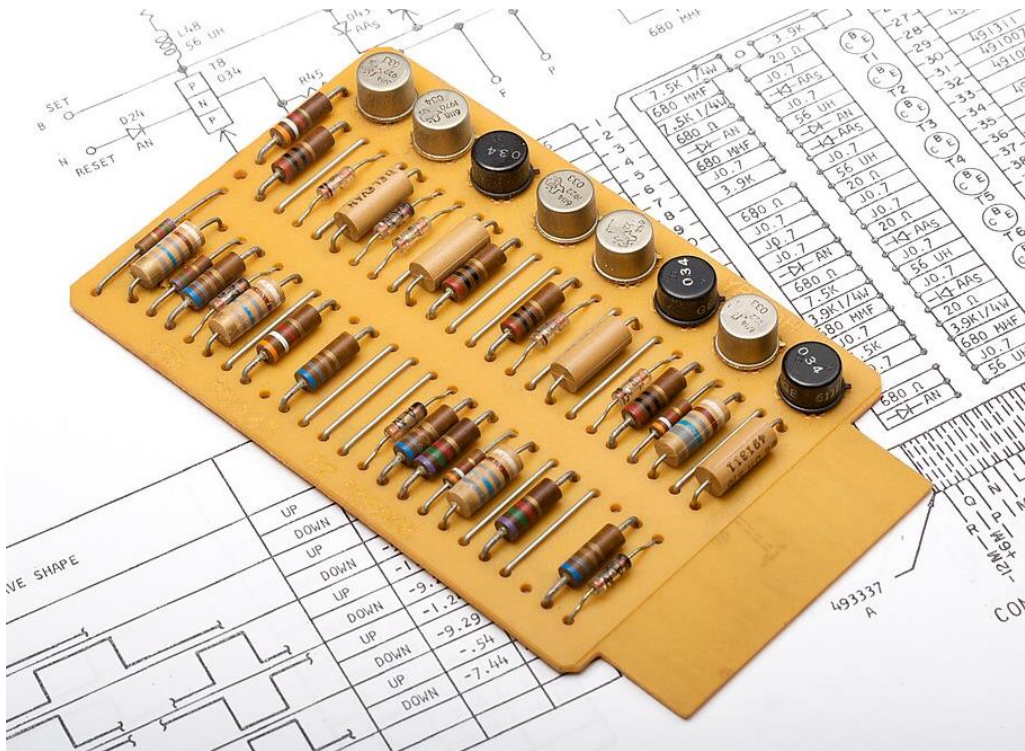
Gordon Moore, co founder of Intel, outlined several contributing factors for this exponential behavior of Semiconductor complexity:

- chip manufacturers could work with larger areas without losing reduction yields; die sizes were increasing at an exponential rate
- simultaneous evolution to finer minimum dimensions;
-- the "circuit and device cleverness"



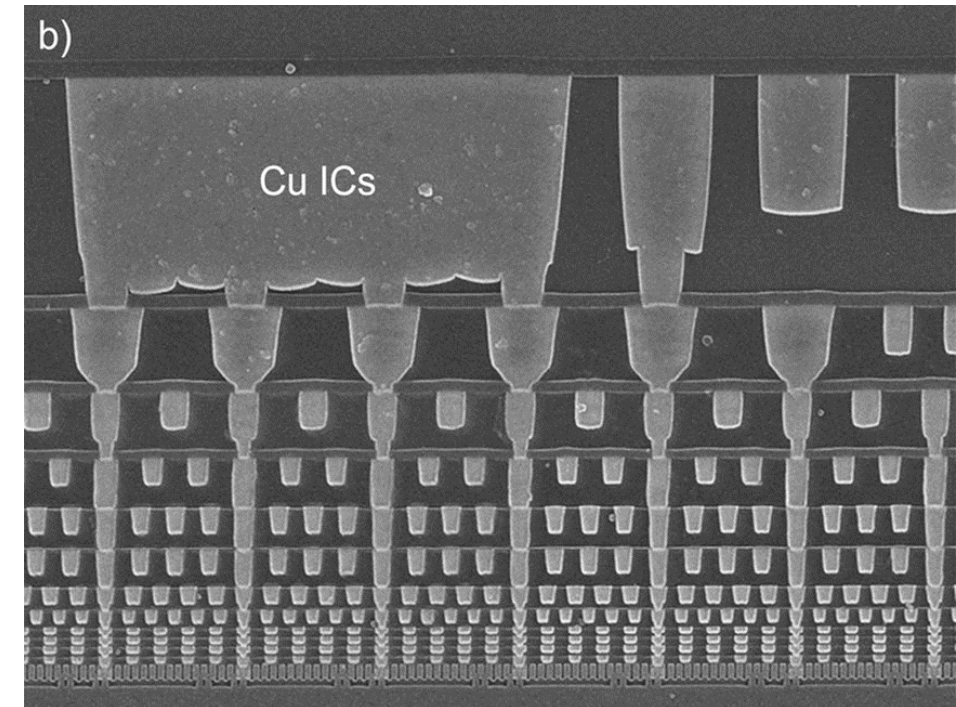
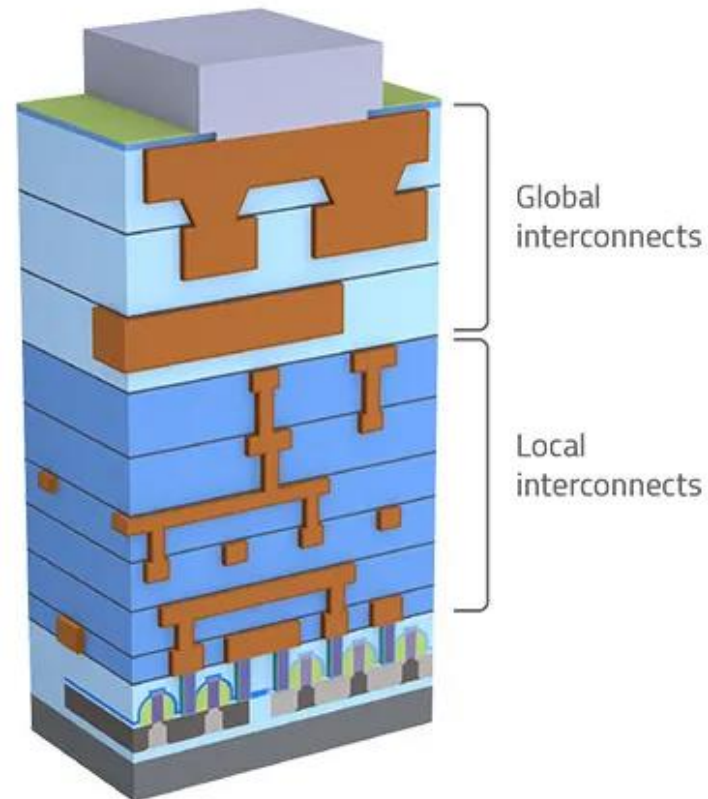
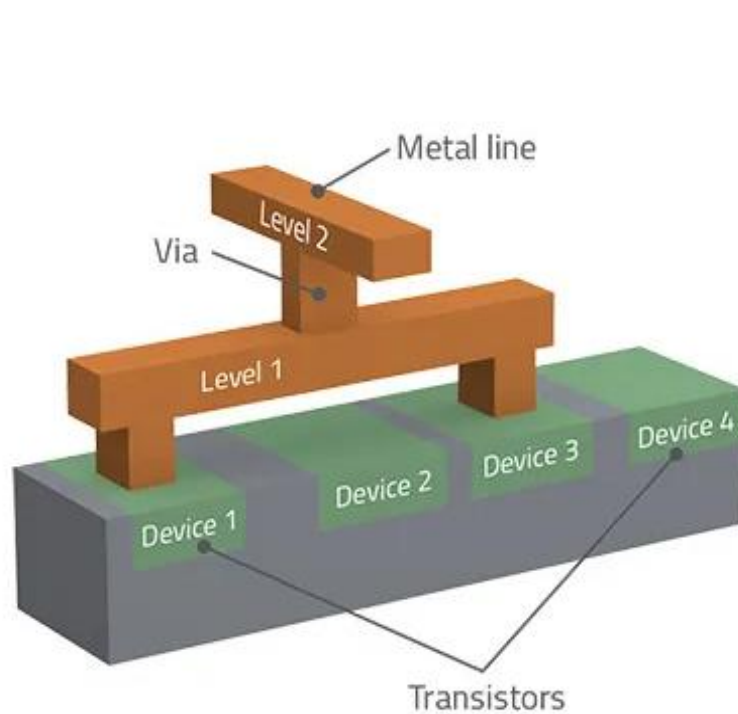
Moore's law is the observation that over the history of computing hardware, the ***number of transistors on integrated circuits doubles approximately every two years to improve the chip performance (being a combination of the effect of more transistors and their being faster).***

Log-linear relationship between device complexity **(higher circuit density at reduced cost)**

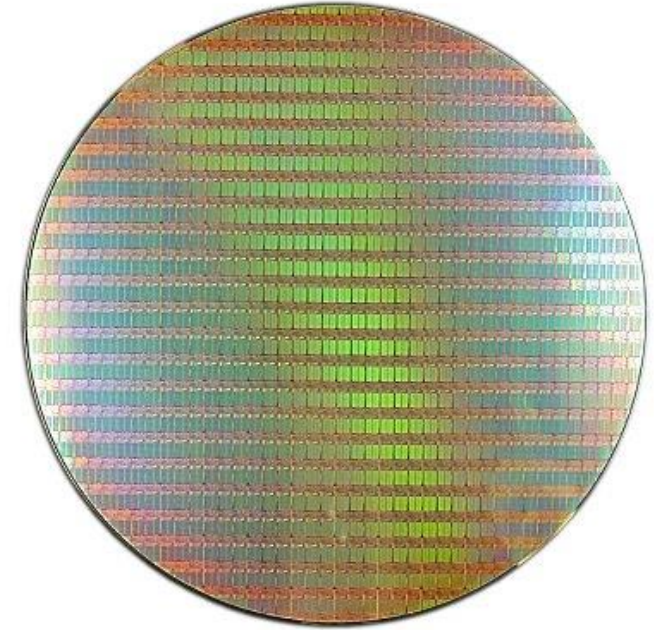


10th generation Intel Core i7 processors have approximately 1.7 billion transistors

Integrated Circuit (IC)

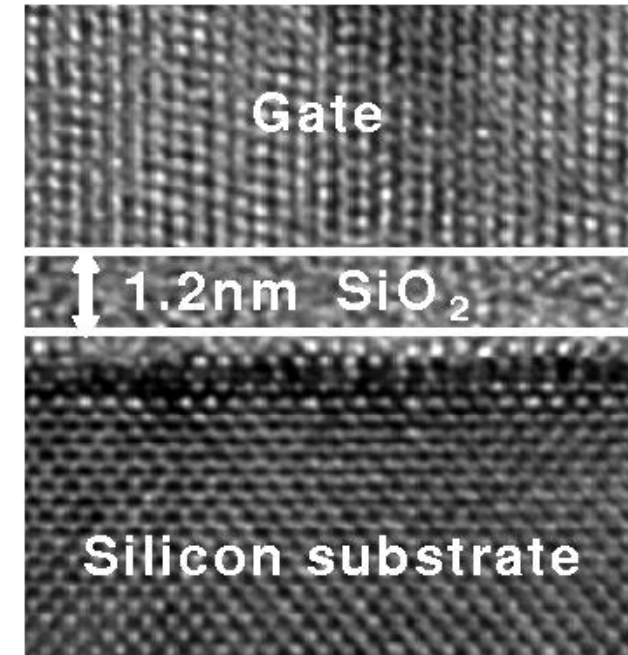
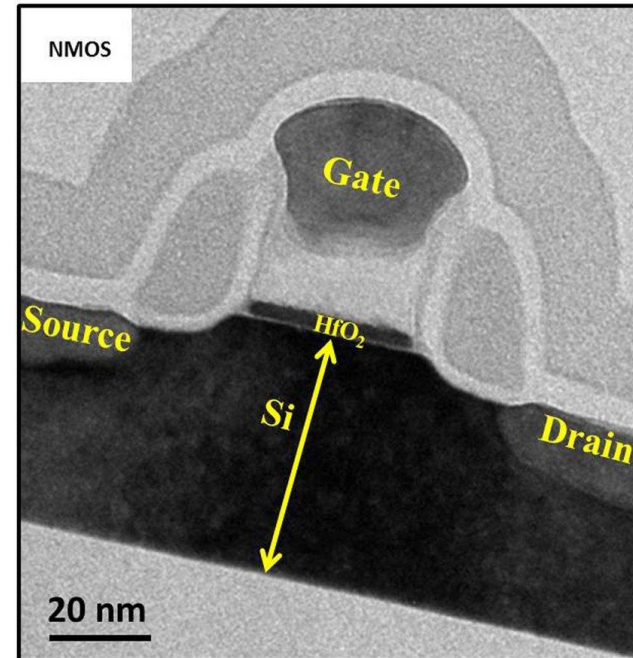
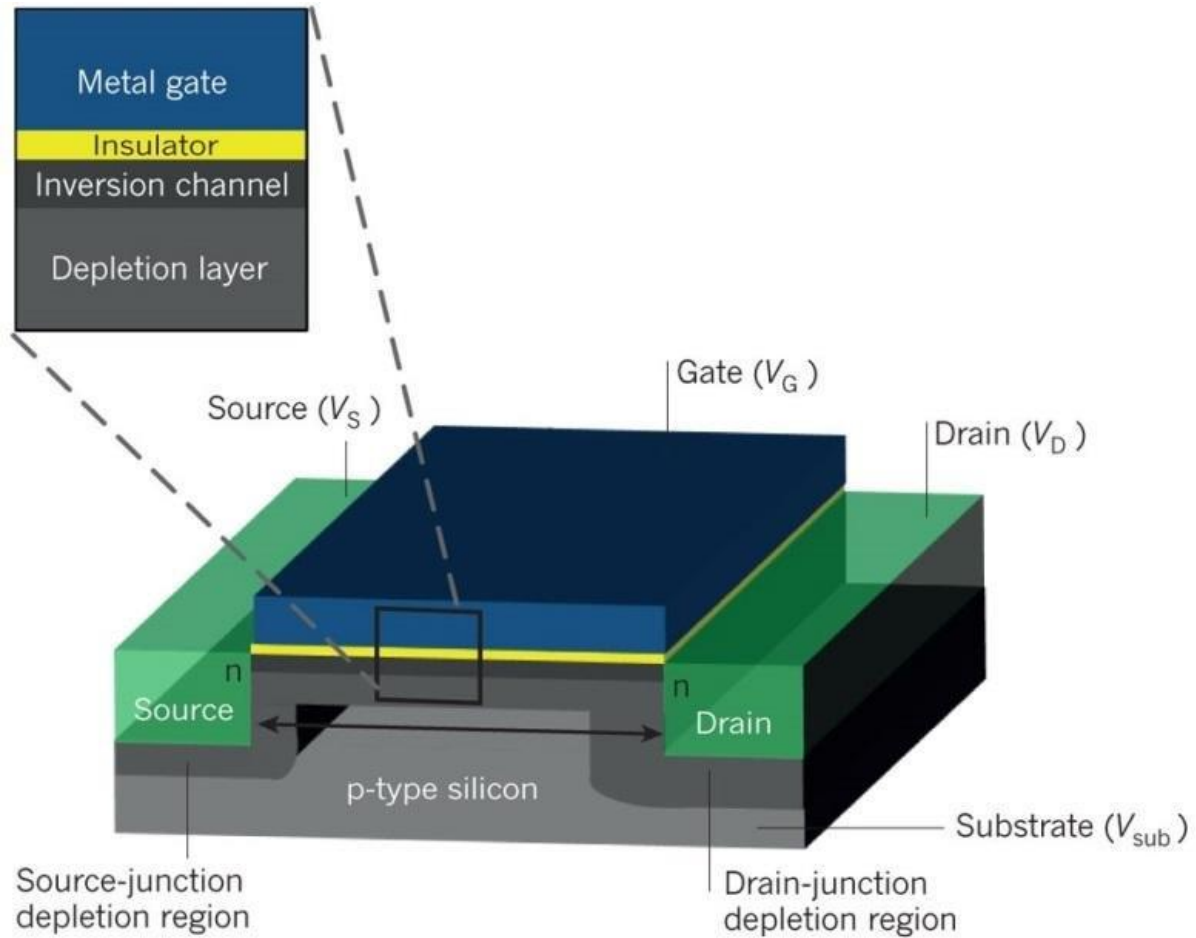


Semiconductor wafer



A wafer is a thin slice of semiconductor, such as a crystalline silicon, used for the fabrication of integrated circuits, solar cells, etc.

A transistor



Ref: <https://doi.org/10.1038/s41598-017-07418-y>

Band Diagrams

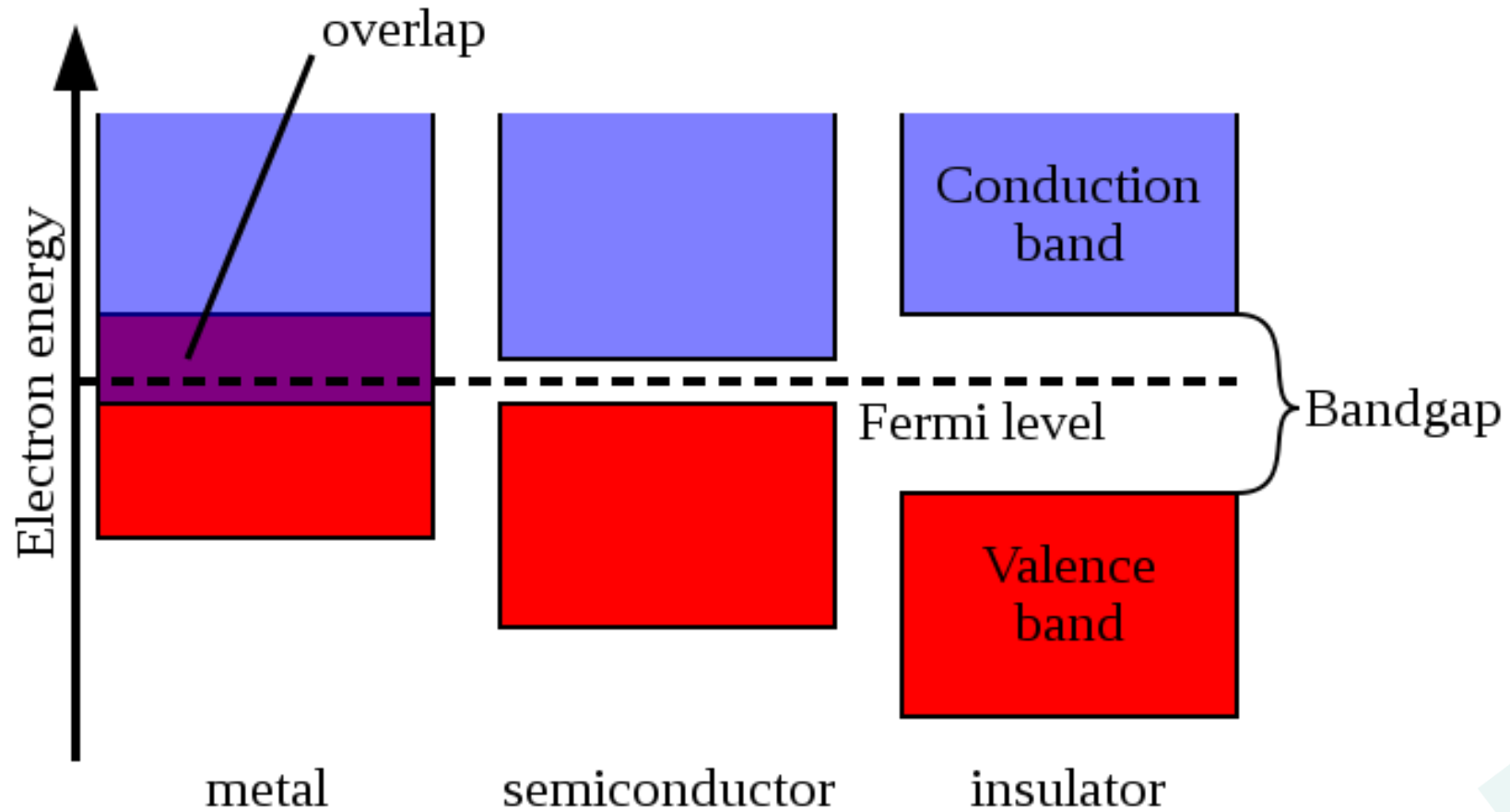


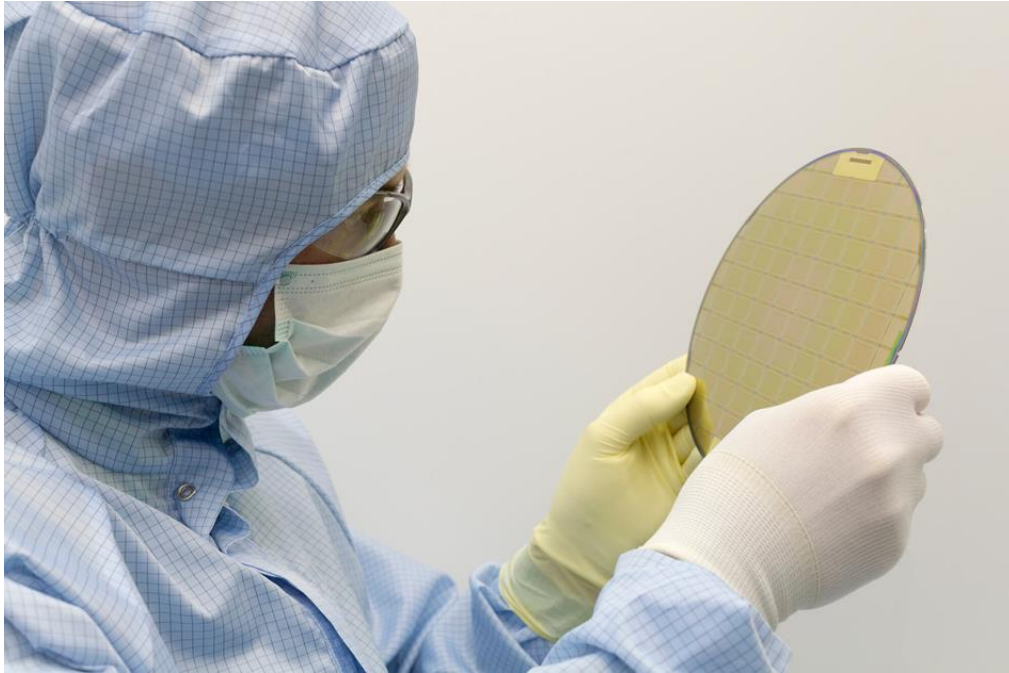
Table 1.1 | A portion of the periodic table

III	IV	V
5 B Boron	6 C Carbon	
13 Al Aluminum	14 Si Silicon	15 P Phosphorus
31 Ga Gallium	32 Ge Germanium	33 As Arsenic
49 In Indium		51 Sb Antimony

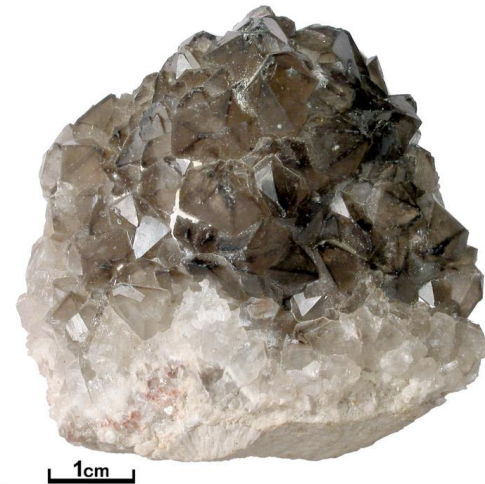
Table 1.2 | A list of some semiconductor materials

Elemental semiconductors	
Si	Silicon
Ge	Germanium
Compound semiconductors	
AlP	Aluminum phosphide
AlAs	Aluminum arsenide
GaP	Gallium phosphide
GaAs	Gallium arsenide
InP	Indium phosphide

Why Silicon?



A scientist, holding a Silicon wafer in microfab. lab



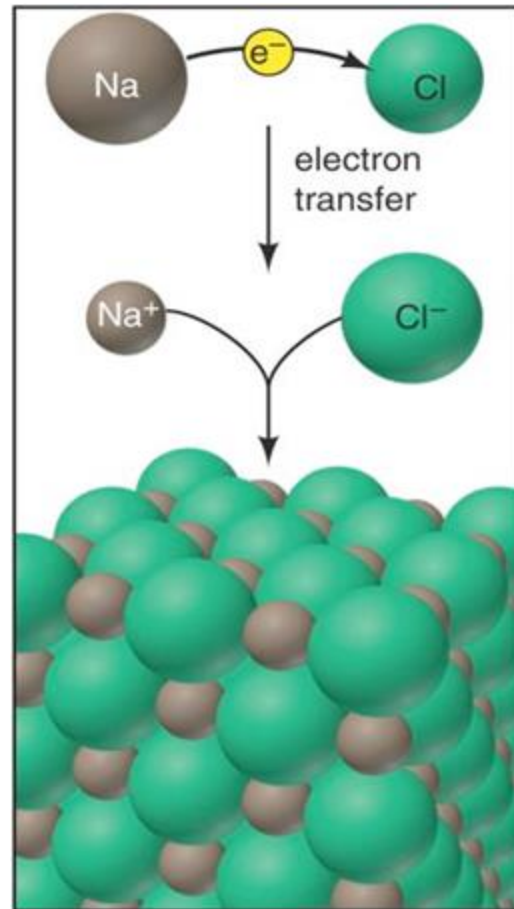
Silicon is a very common element, the main element in sand & quartz.

Only “Easy availability”? Or Anything else!

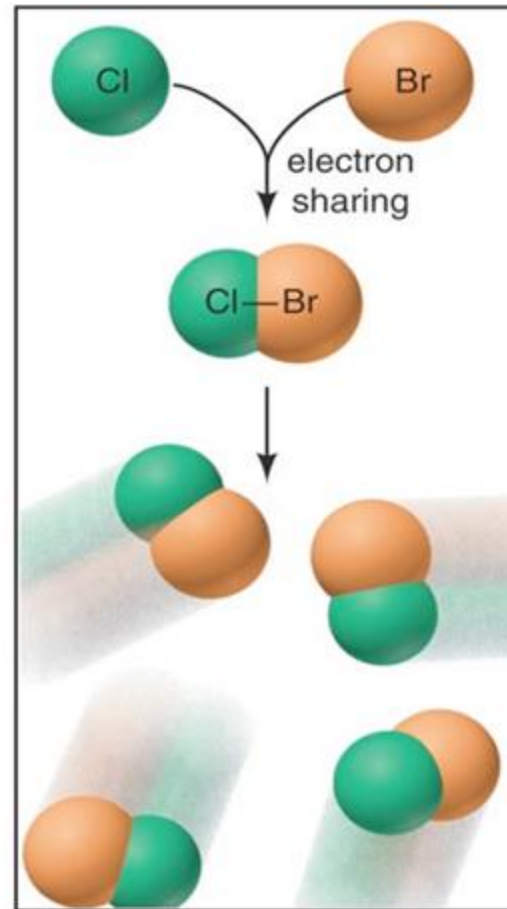
Chemical bonding



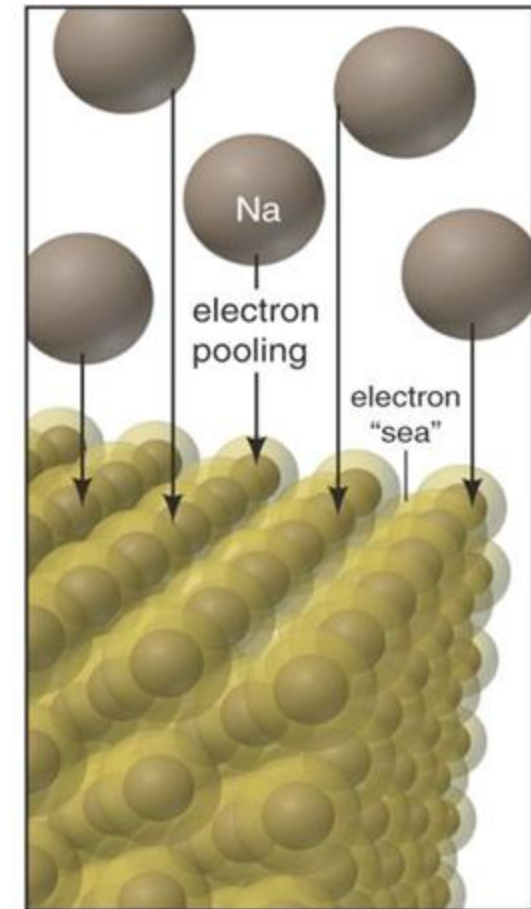
Types of Chemical Bonds



A Ionic bonding



B Covalent bonding



C Metallic bonding

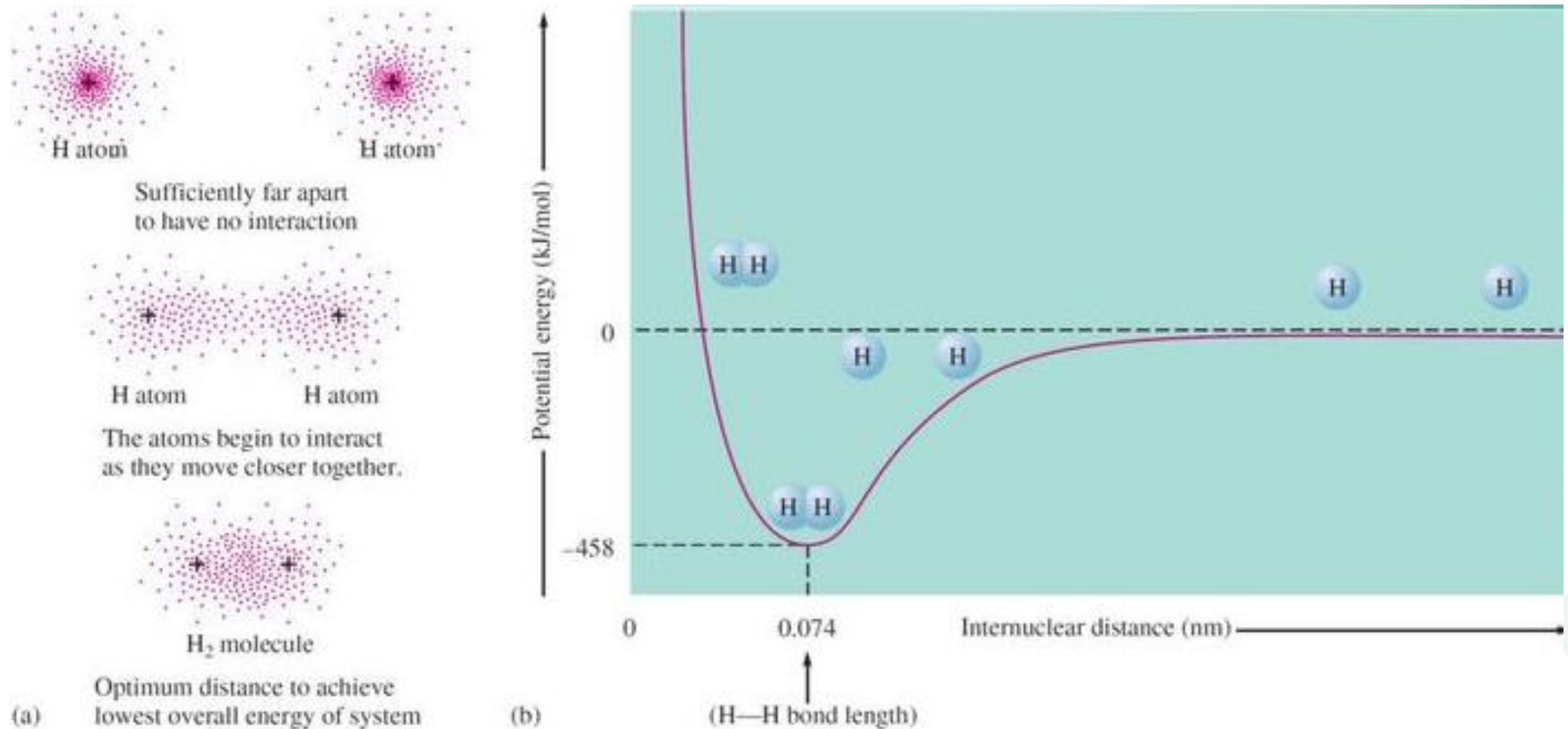
Bonding Basics



Atoms will position themselves so that the system will achieve the lowest possible energy.

- **The distance where this energy is minimal is the bond length** (distance between 2 nuclei, usually in nm).
- The energy required to break a bond is called the **bond energy** (usually in kJ/mol).
- ***The shorter the bond length (the stronger the attraction and the higher the bond energy.)***

Interaction of Hydrogen atoms and energy profile



The **bond length** is the distance at which the system has minimal energy

Questions



- Rank in order of increasing bond energy:



Answer

- With increase in atomic size from F to Br bond length increases
- As the **bond length increases** the **bond energy decreases**

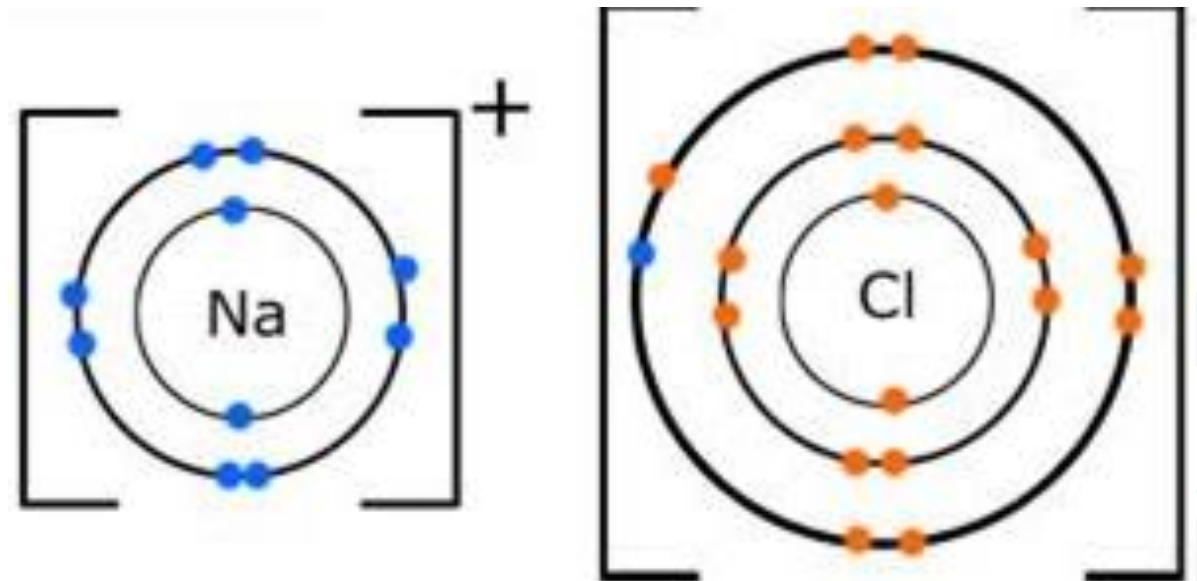
In order of increasing bond energies.



Ionic Bonds



- An attraction between anions and cations
- Electrons are completely transferred.
- A bond between a metal and a nonmetal.
- Stronger than covalent bonds.
- Commonly called metallic salts.



Ions Formation



- **Metal atoms** tend to lose all valence electrons to form cations. Cations are smaller than parent atom
- **Nonmetal atoms** tend to gain enough electrons to fill the outer s and p subshells. Anions are larger than parent atom
- **Ionic Radius:** The Ionic radius can be described as the distance between the nucleus of an ion and the outermost shell of the ion.
(What is then atomic radius?)
- **Isoelectric ions-** ions containing the same number of electrons...the size decreases as the nuclear charge increases.

Ionization Energy



- The energy needed to remove one or more electrons from a neutral atom to form a positively charged ion.
- By definition, the first ionization energy of an element is the energy needed to remove the outermost, or highest energy, electron from a neutral atom in the gas phase.



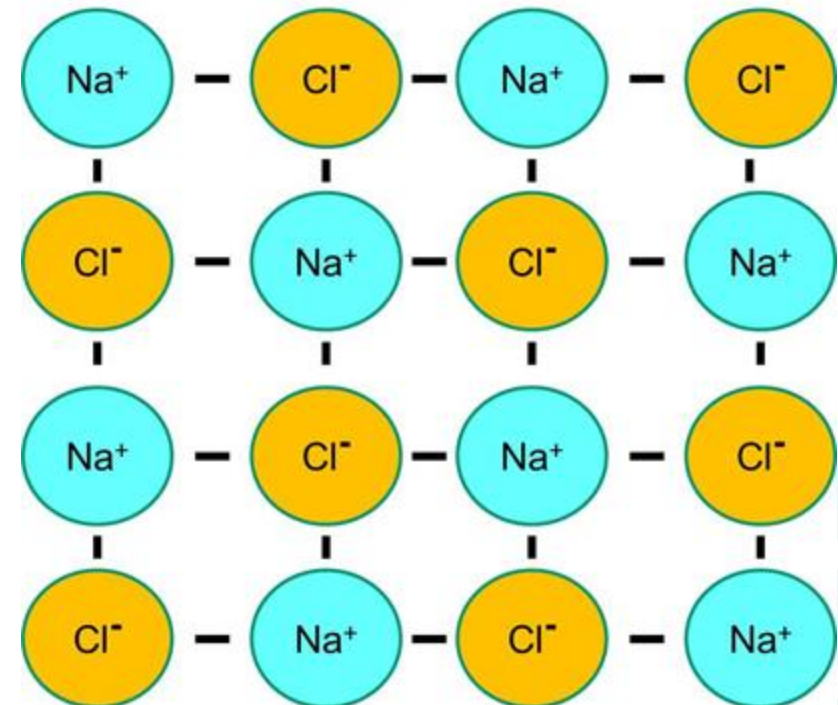
Properties of Ionic compounds



- High MP & BP
- Hard, not easily crushed
- Conduct electricity when melted or dissolved because ions are freed up to move.

- The ions are in a very strong CRYSTAL LATTICE pattern.

Crystal Lattice of NaCl (table salt)



Coulomb's law can be used to calculate the energy of an interaction between a pair of ions. It can be used for both attractive and repulsive forces.

$$F_e = k_e \frac{q_1 q_2}{r^2}$$

where

- F_e is the force
- k_e is the Coulomb's constant ($8.987 \times 10^9 \text{ N.m}^2.\text{C}^{-2}$)
- q_1 and q_2 are the signed magnitudes of the charges
- r is the distance between the charges

- The **strength of an ionic bond** can be determined by 2 factors:
 - **The charge of the ions**, the higher the numerical charge value the stronger the attraction.
 - **The distance of the ions**, the closer they are the stronger the bond.

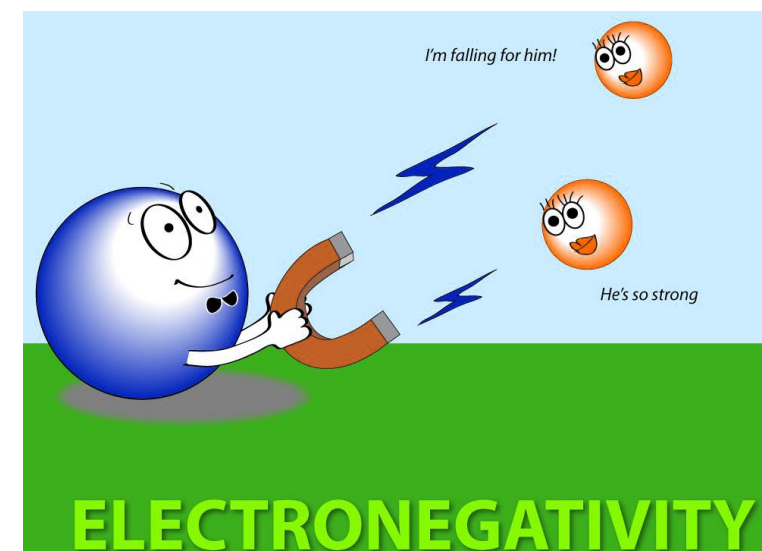
Covalent Bonds



Just in case you forgot

- This type of bonding most occurs between a **nonmetal and a nonmetal**, or between **metalloids and nonmetals**.
- Lower MP & BP
- Tend to be volatile gases or liquids
- Softer substances and crush easily
- Poor conductors of electricity

- **Electronegativity** is the attraction an atom has for a shared pair of electrons.
- Electronegativity increases across a row and up a column

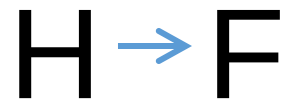


Covalent Bonding

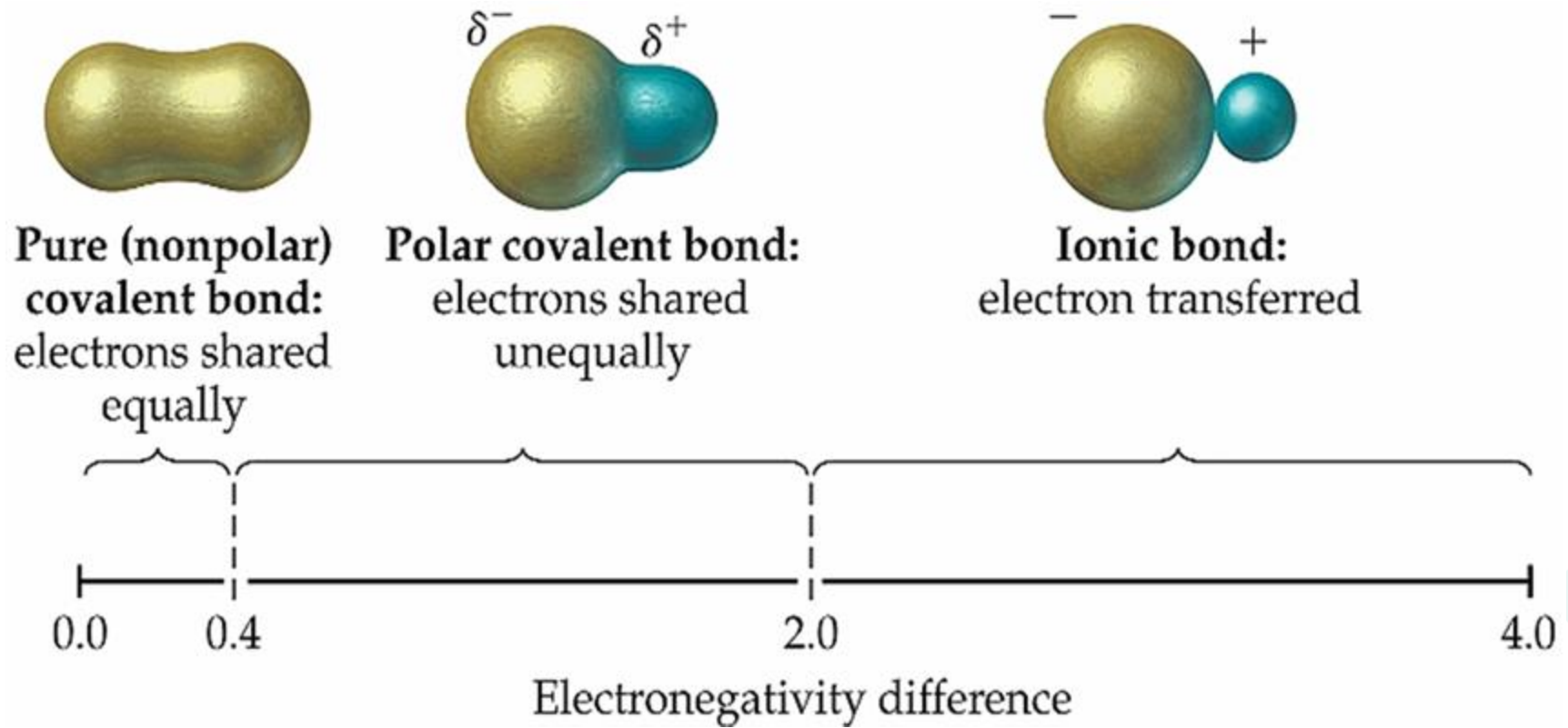


When two atoms have a small difference in their tendencies to lose or gain electrons, we observe electron sharing and covalent bonding.

- Differences in electronegativity can be used to determine if the electrons are shared equally or unequally.
- Often, instead of using a single line to indicate the covalent bond, **an arrow is used with the head pointing toward the atom with the greater attraction for the electrons.**



Types of Covalent Bonds



Lattice Energy



- The energy required to break an ionic bond into its gaseous foundational elements:



Lattice energy increases as:

- Ionic size decreases
- Ionic charge increases

As the size of lattice energy increases, it becomes difficult for ions to get separated from the ionic lattice which gives more stability to the ionic solid.

Therefore, ionic solids do not convert into gaseous state (sublime) at room temperature and they melt at high temperatures

Covalent Bond Energies



- Breaking bonds requires energy, and the process is endothermic.
- Based on the type of atoms involved in the bond, we can use existing tables to look up the energy required to break the bond.
- The values will be in kJ/mol

Table 8.5 ► Bond Lengths for Selected Bonds

Bond	Bond Type	Bond Length (pm)	Bond Energy (kJ/mol)
C—C	Single	154	347
C=C	Double	134	614
C≡C	Triple	120	839
C—O	Single	143	358
C=O	Double	123	745
C—N	Single	143	305
C=N	Double	138	615
C≡N	Triple	116	891

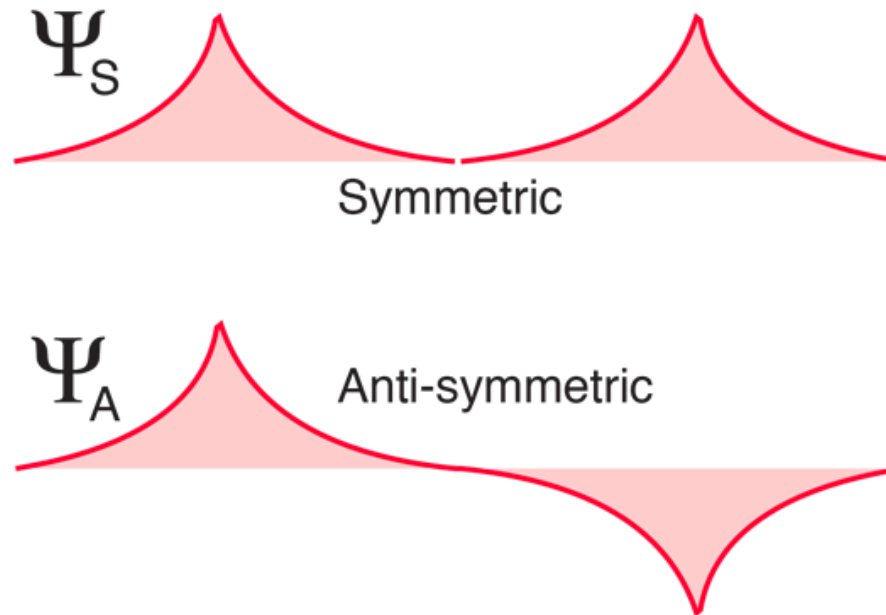
Relationships:



- Atomic radius will influence bond length, larger atoms will have longer bond lengths.
- Bond length and bond energy are inverse, as the bond length increases the bond energy decreases.
- As the number of shared pairs of electrons increases, bond length shortens and bond energy increases.

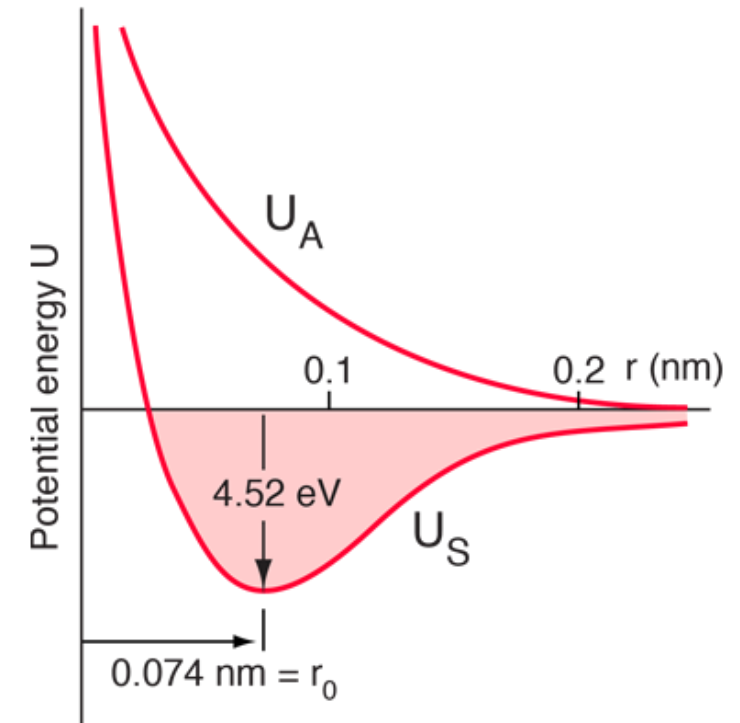
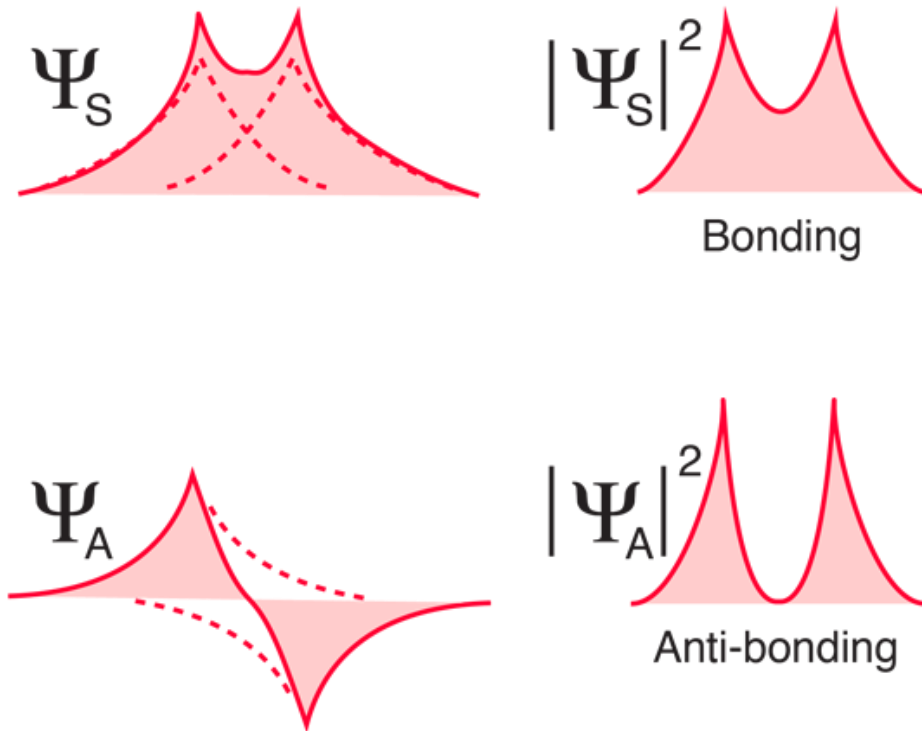


Hydrogen Molecule



- The electron distribution around the protons of the hydrogen is described by a quantum mechanical wavefunction.
- The wavefunction which describes the two electrons for a pair of atoms can be symmetric or antisymmetric with respect to exchange of the identical electrons.
- From the Pauli exclusion principle, we know that the wavefunctions for two identical fermions must be antisymmetric. The electron spin part of the wavefunction can be symmetric (parallel spins) or antisymmetric (opposite spins), but then the space part of the wavefunction must be the opposite. That guarantees that the entire wavefunction (the product of the spin and space wavefunctions) is antisymmetric.

Contd...



- As shown, when the hydrogen atoms are brought close together the symmetric spatial wavefunction leads to a bonding configuration of electrons and the antisymmetric one does not.
- The actual electron charge density is given by the square of the magnitude of the wavefunction, and it can be seen that the symmetric wavefunction gives a high electron density between the nuclei, leading to a net attractive force between the atoms (a bond).